Cover Page

BUILT ENVIRONMENT
CASE COMPETITION (BECC) 2024

Feasible and Cost-effective Solutions to Safeguard Singapore's Northwest Coast

<Roller Coastal>

1. Introduction

To combat the rising sea levels and protect the Northwest coast, Singapore requires substantial financial resources. The government has emphasised the need for around $100 billion [1] over the long term, with an initial $5 billion fund [2] established in 2021 to kickstart these efforts. In 2022, PUB launched RFP 2201 [3], focusing on using emerging technologies to support coastal protection. This initiative offers 100% funding for approved costs and 30% overheads for public sector entities and IHLs [4]. The importance of these efforts is underscored by the critical role of the Western Water Catchment, one of the two main national water catchments and the only plant supplying treated reservoir water to the western part. Besides, the biodiversity-rich SBNPN is home to flora & fauna [5]. However, the region faces significant challenges, such as the proximity of the border to the coastline and its severe land scarcity caused by Singapore's high population density. These issues necessitate innovative and efficient solutions to avoid border conflicts and optimise land use. Thus, we aim to propose cost-effective and feasible strategies to safeguard this region.

2. Northwest Coast 1

2.1 Site Analysis

Northwest Coast 1 is a 24 km stretch from Tuas checkpoint to Lim Chu Kang jetty. The coastline protects four coastal reservoirs, which were created by damming up river mouths, making the dams, dikes, and sluice gates part of the coastal barrier. Moreover, the coastline is protected by a perimeter of mangroves[6](Fig. 1.), and is quite inaccessible to the public as it is also Singapore's security barrier at the Johor Straits.

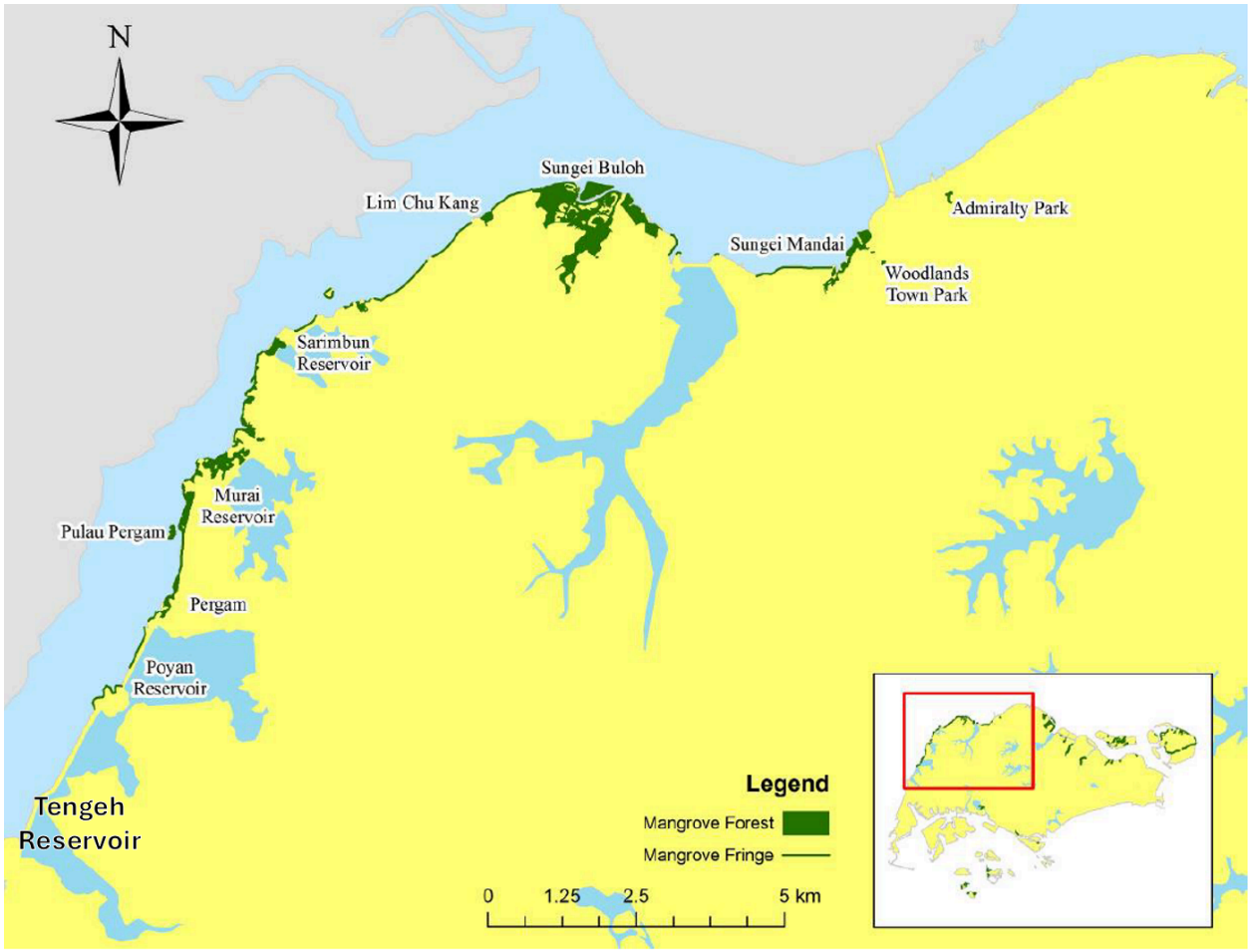


Fig. 1. Mangroves along coastline [6]

2.2 Solution

Building upon Tengeh reservoir's current solar farm, this stretch can be developed into an energy hub to serve Singapore's sustainable efforts. Therefore, our solution aims to safeguard our coastline with the sub-goal of sustainable energy development.

2.2.1 Solar Panel Integrated Floating Breakwater

Off coast near Tengeh reservoir, an array of hybrid floating breakwater(Fig. 2.) can be deployed along the coast. This can effectively dissipate wave energy, while providing additional energy through the integrated solar farm[7]. Floating breakwaters are easier to move and adjust, making them flexible for changing needs and are less harmful to marine life compared to fixed ones. This is an applicable way to expand the current existing Tengeh Reservoir solar farm.

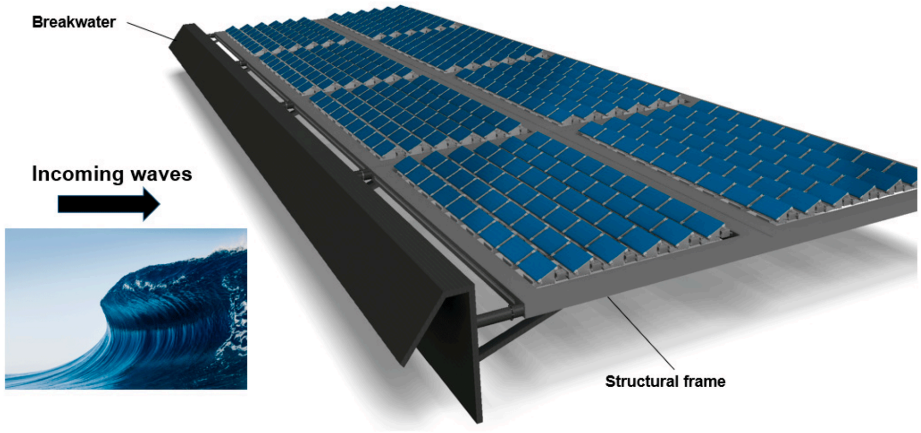


Fig. 2. Solar Panel Integrated Floating Breakwater

2.2.2 Mangrove expansion

Mangroves are nature's best breakwater, as they have the combined function of desalination, wave barrier, and storing CO_2. Moreover, underground seawater intrusion is more detrimental than waves overtopping[8], and mangroves can desalinate the underground seawater to safeguard the coastal reservoirs. Therefore, we propose to expand the current mangrove coastline to increase its effectiveness.

As early phases of propagule growth is difficult, TetraPOT [9] can be used for early stage protection while the seedlings grow within the tetrapods. This cost effective implementation is easy to adapt for its simple construction process.



Fig. 3. TetraPOT

2.2.3 Xbloc Revetments and Ecological Enhancement

Concurrently, Xbloc can be adapted along the coastline. Xbloc is a sustainable and economical material that can be used as dikes to protect the coast road[10], which may be progressively raised in the future. To further reinforce the coastline that makes up part of the reservoirs, ecological enhancement is recommended. By adding on or modifying the textures of the revetments, it allows an environment for organisms to grow. This enhances the wave attenuation, erosion protection, and biodiversity of the area.



Fig. 4. Xbloc units [9]

To summarise, the proposed solution is illustrated in Fig. 5 and Fig. 6.

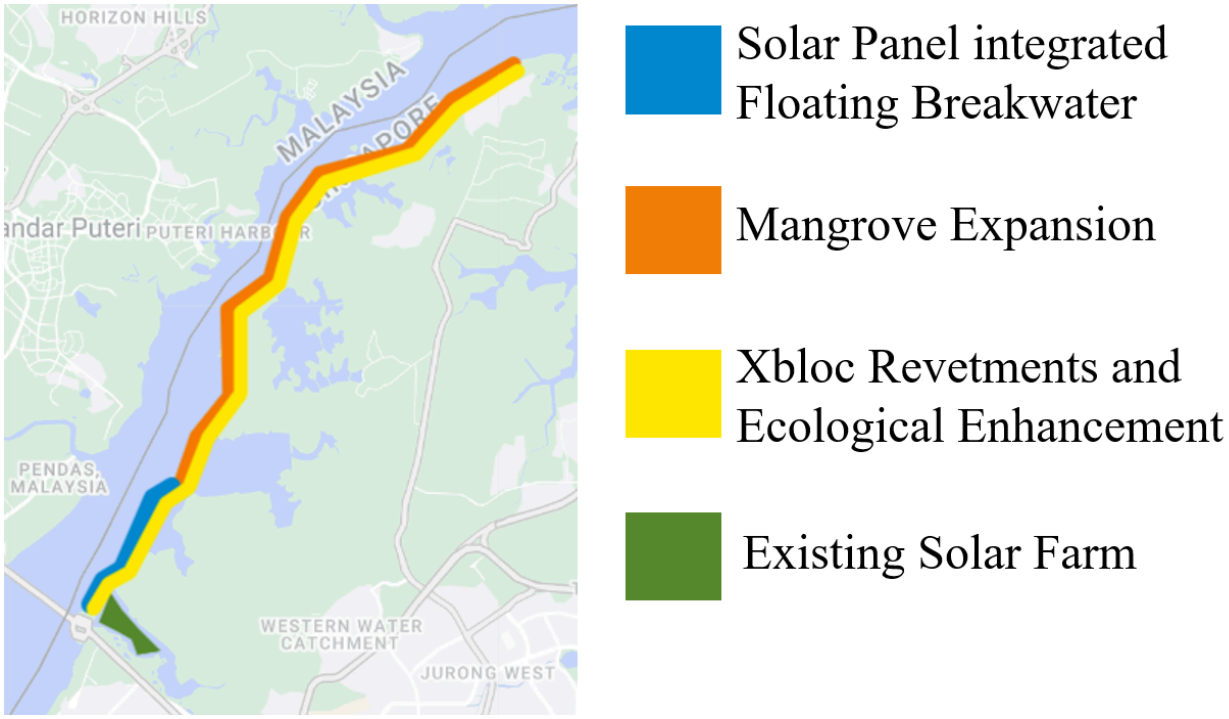


Fig. 5. Northwest Coast 1 Solution Plan View

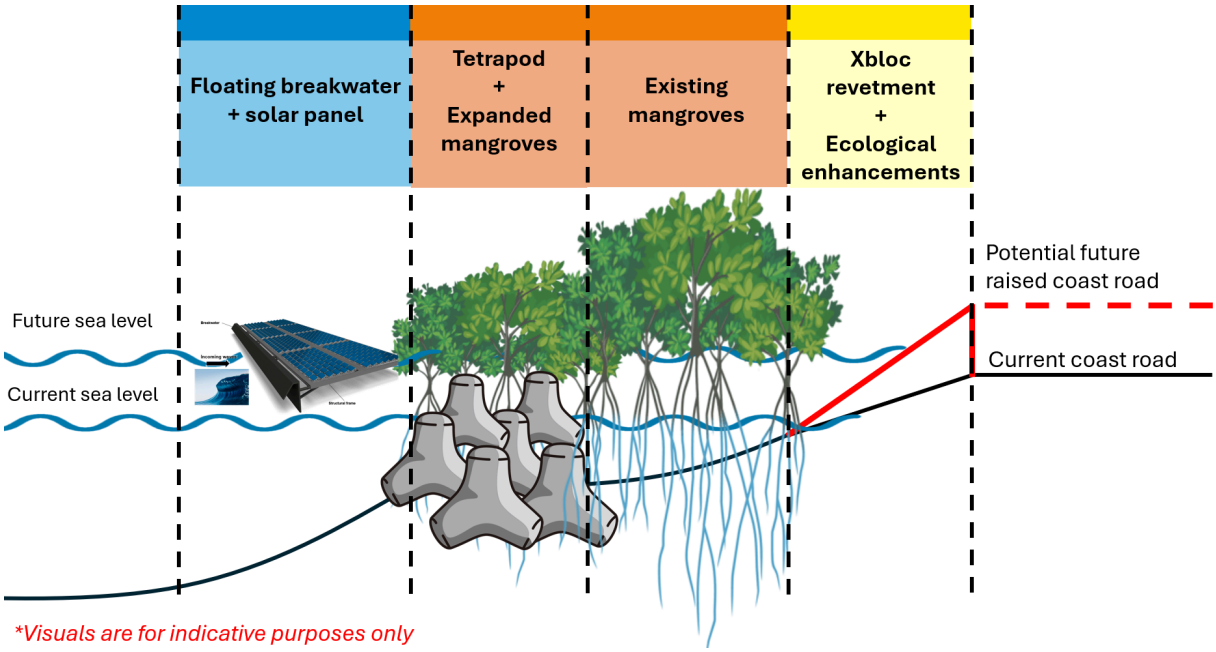


Fig. 6. Typical Cross Section of Solution at Northwest Coast 1

3. Northwest Coast 2

3.1 Site Analysis

At Site C2, Kranji Reservoir, Kranji Marshes and Sungei Buloh Wetland Reserve are some critical ecological zones in Singapore. Currently, the farmlands near Kranji Reservoir are at risk of soil salinization[11]. The shoreline in C2 also has a medium level of erosion. Furthermore, due to the Kranji Reservoir being directly connected to the coast and underground saltwater intrusion, it is also among the areas that face the highest level of salinization vulnerability in Singapore [11]. Furthermore, the future development in the region adds threats to the coastal resilience of the coast. Surrounding them are Kranji Industrial Zones industrial areas and upcoming major developments such as the Woodlands Checkpoint expansion [12]. These lead to coastal erosion, loss of intertidal habitats, and changes in tidal patterns.



Fig. 7. Upcoming Woodlands Checkpoint Expansion

The worrying seawater level rise, especially at low-lying areas, such as Sungei Buloh, will also see higher frequency of floods during high tide [13]. Coupling with the upcoming developments such as Lim Chu Kang Urban Agrifood Cluster and Sungei Kadut Ecodistrict, mitigating the impacts at this site has become extremely important and it requires holistic planning and implementation to safeguard this area's coastline [14].

3.2 Solution

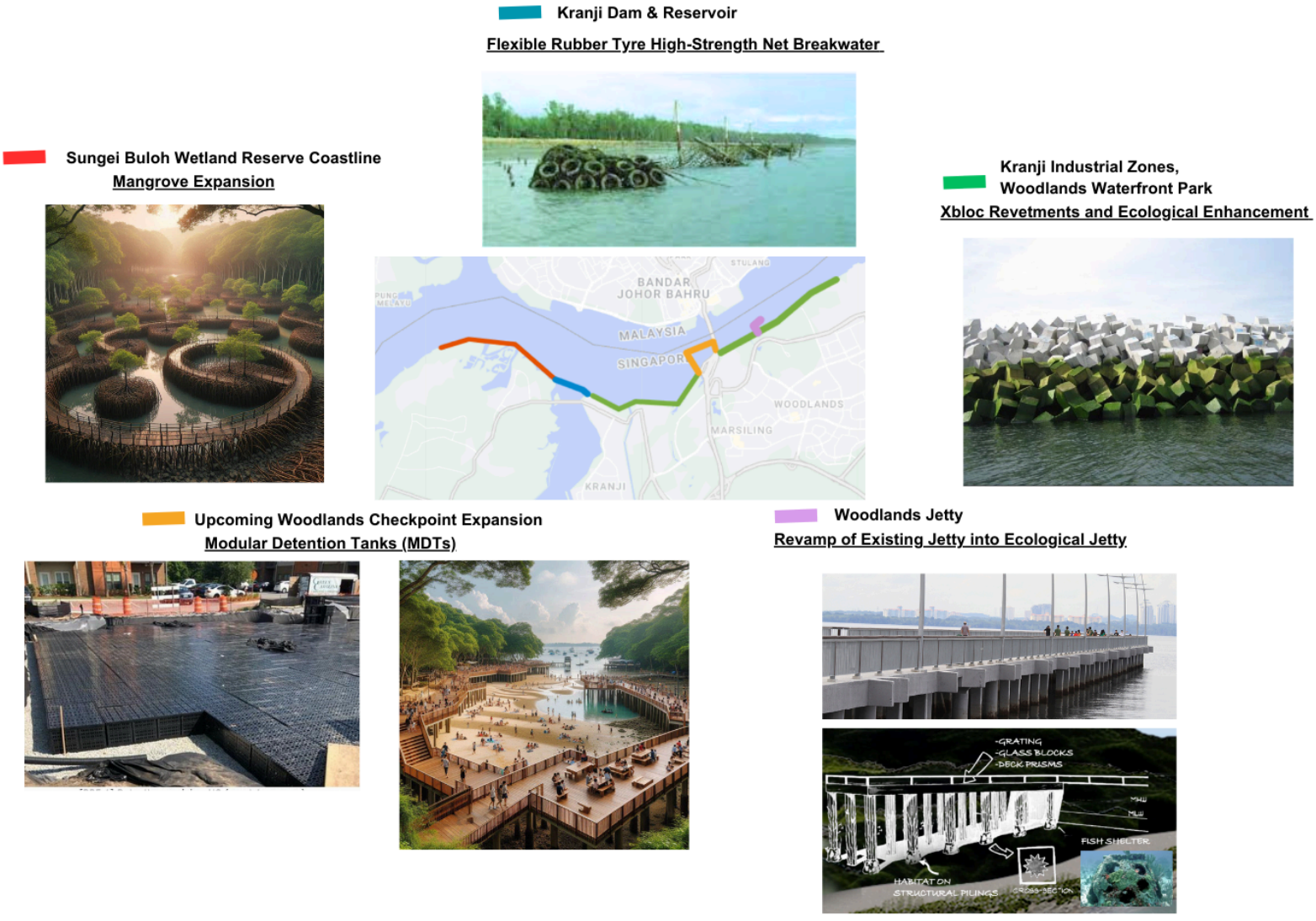


Fig. 8. Various Plans for Site C2

Similar to Coast 1 above, the mangrove at Sungei Buloh Wetland Reserve Mangrove will see expansion in its size and species.

3.2.1 Flexible Rubber Tyre High-Strength Net Breakwater

This solution will be implemented at Kranji Reservoir, which will consist predominantly of concrete posts, triangular gabions filled with stones, and flexible high-strength nets. The high-strength nets, such as wire/nylon nets, are used to reduce wave energy and to help sediment deposition. These nets are placed in between two triangular gabions. They will be filled with used vehicle tires with wire fins (with diameters of 0.58m), which act to hold the flexible nets with the help of concrete posts, and to reduce wave energy and increase roughness.

3.2.2 Modular Detention Tanks (MDTs)

At the upcoming new Woodlands Checkpoint, we will use MDTs. MDTs are pre-fabricated tanks that are ideal for land reclamation projects. Placed beneath the reclaimed area, MDTs will capture rainwater runoff from the checkpoint to prevent flooding. We aim to store 1.5million m^3 water.

3.2.3 Revamp of Existing Jetty into Ecological Jetty

We aim to replace traditional docks and piers by incorporating improved lighting and structured pilings. Fibreglass grating and reflective panels can be utilised. By coupling technologies that allow light penetration and reflectivity, optimal light conditions are created, promoting the growth of algae seagrass, and enhancing nursery habitats for fish and invertebrates. Textured pilings and fish shelters integrated into the dock design increase intertidal habitat area, benefiting fish and vegetation.

3.2.4 Xbloc Revetments and Ecological Enhancement

(Refer to Section 2.2.3)

4. Estimated Initial Cost for Projects

Table 1. Initial Cost for Northwest Coast 1

Items	Cost per unit (SGD)	Quantity	Total Cost(SGD)
TetraPOT	384.62	28,600	11,000,000
Mangrove Tree Propagule	3.50	28,600	100,000
Floating breakwater (per 25m)	10,000	160	1,600,000
Solar panel (per 7000 m^2)	65,000	50	3,250,000
Xbloc revetment (per running m)	2,000	24,000	48,000,000
Ecological enhancement (per running metre)	2,000	24,000	48,000,000
TOTAL			111,950,000

Table 2. Initial Cost for Northwest Coast 2

Items	Cost per unit (SGD)	Quantity	Total Cost(SGD)
TetraPOT	384.62	5720	2,200,026.40
Mangrove Tree Propagule	3.50	5720	20,020
Tyre High Strength Net Breakwater (per square metre)	$40	5000	200,000
Modular Detention Tank (per cubic metre)	85	1,500,000	127,500,000
Xbloc revetment (per running metre)	2000	4000	8,000,000
Ecological enhancement (per running metre)	2000	4000	8,000,000
Ecological Jetty Renovation fee	600,000	1	600,000
TOTAL			146,520,046.40

4. Impact Assessment (Triple Bottom Line)

4.1 Social

The project is effective to combat coastal erosion and seawater intrusion which will damage the coastal reservoir, further threatening drinking water supply of western part. The enhanced shoreline with its improved biodiversity and scenic floating wetlands will provide new recreational opportunities and educational resources.

4.2 Economy

Our measures are more cost-effective compared to traditional solutions. They reduce needs for expensive repairs and the increased lifespan of coastal protection infrastructure. Solar panel integrated floating breakwater will provide reliable clean energy, maximising resource utilisation in land-scarce Singapore.

4.3 Environment

The expansion of mangroves along with the Xbloc revetment will create new habitats for various marine and terrestrial species. The mangrove forest will mitigate urban heat stress, prevent seawater intrusion to freshwater habitat and act as natural barriers against strong waves.

5. Sustainability Quantification

We have referred to Life Cycle Assessment (LCA) [15], and defined the sustainability coefficient as in Fig. 9. [17]

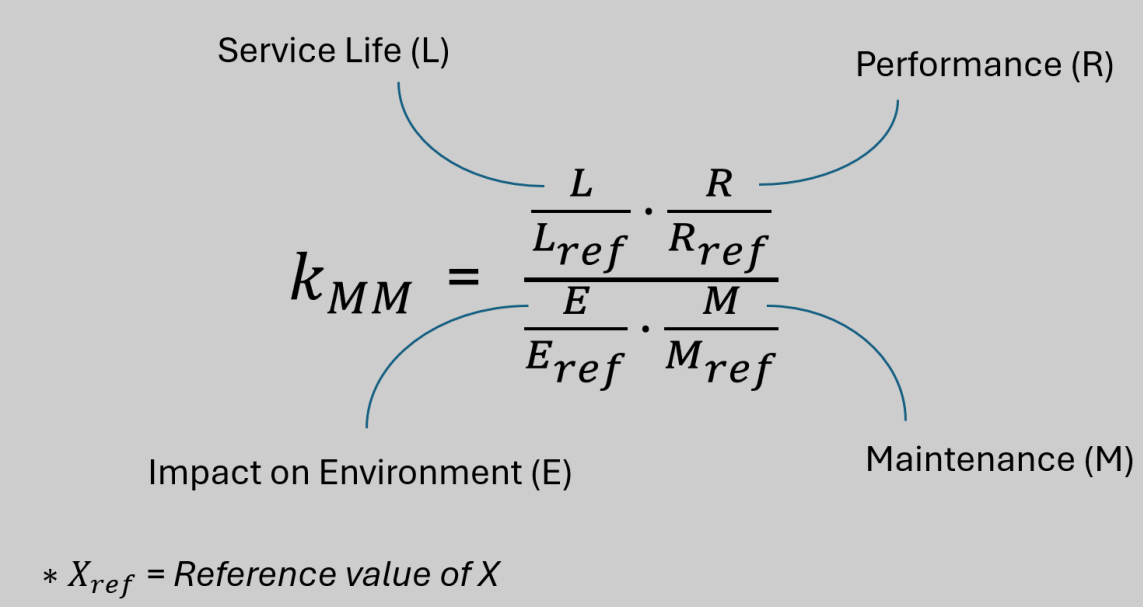


Fig. 9. Expression for Sustainability Coefficient k_{MM}

Table 3. Initial Cost of Materials

	Reference	**Sustainable**
Tetrapot	300 SGD	400 SGD
Floating Breakwater	400000 SGD	600000 SGD
Bio Revetment	300000 SGD	400000 SGD

Table 4. shows that all the sustainability coefficients are larger than 1, indicating our solutions are more sustainable compared to traditional ones. Service Life and Performance Ratio shows sustainable materials perform slightly better while Environment Ratio indicates that sustainable materials significance reduces negative environmental impacts.

Table 4. Sustainability Coefficient Table

	Tetrapod	**Floating Breakwater**	**Bio Revetment**
Sustainability Coefficient K_{MM}	1.123	3.66	2.16
Service Life Ratio (L/L_{ref}) (Refer to Table 6 in Appendix)	0.97	0.97	0.67
Performance Ratio (R/R_{ref}) (Refer to Table 7 in Appendix)	1.019	1.019	1
Environment Ratio (E/E_{ref}) (Refer to Table 5 in Appendix)	0.88	0.27	0.31
Maintenance Ratio (M/M_{ref})	1	1	1

**The maintenance cost of both ordinary and sustainable product do not have significant difference

6. Conclusion

In conclusion, our proposed solution emphasises the use of cost-effective measures such as tetrapods, tyre net breakwaters and MDTs which are both economical and sustainable. Additionally, the integration of solar farms can offer a return on investment by generating electricity. Building upon the existing infrastructure, this plan will benefit Singapore in terms of sustainability.

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Appendix

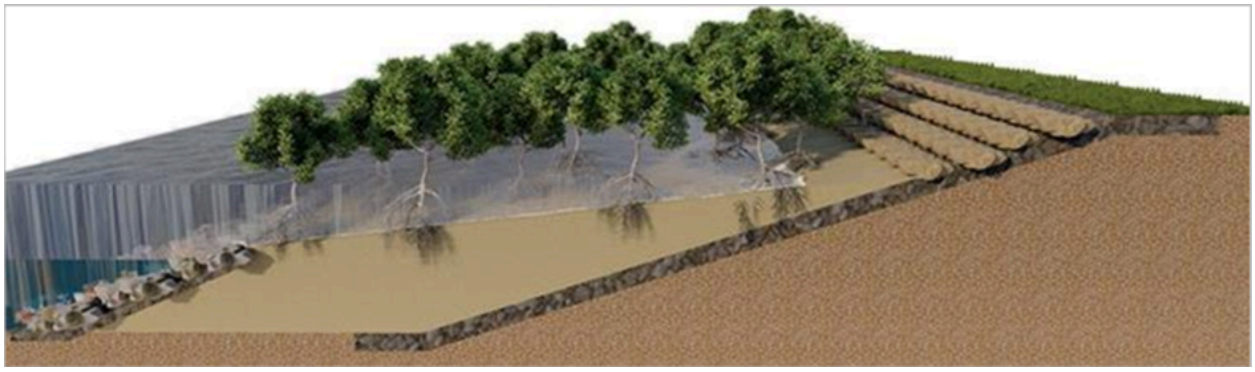


Fig. 10. Hybrid coastal protection system consisting of nature-based feature (e.g. mangrove) and hard structure (e.g. concrete armour)

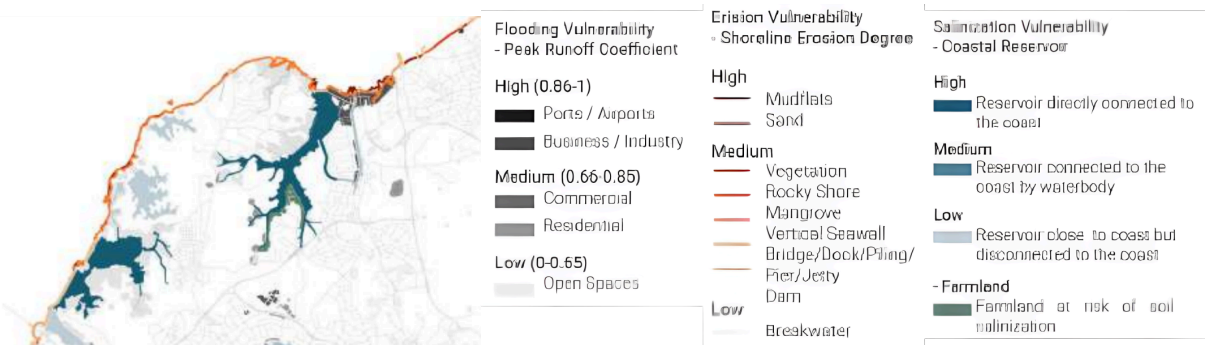


Fig. 11. Northwest Coast Conditions

Referring to the above infographic, As Kranji Reservoir is subject to high vulnerability of erosion, we will adopt Flexible High-Strength Net Breakwater Method.

Some benefits of Floating Tyre High Strength Net Breakwater:

- Lower capital cost, since the main item is the tyres themselves are scrap which are widely available. In Singapore, around 26,000 tonnes of scrap tyre waste is generated annually.

- Simple and quick to build using unskilled labour and a minimum of mechanical equipment - similarly mooring and any repairs are easily achieved with minimum interruption to the protection afforded
- Ecologically safe with respect to

    i) toxicity - tyres are completely non-toxic in seawater, and do in fact attract marine life to the extent that they have been used as submerged reefs and fishing islands,

    ii) impedance of tidal currents which control sedimentation patterns and the regime of estuaries etc.

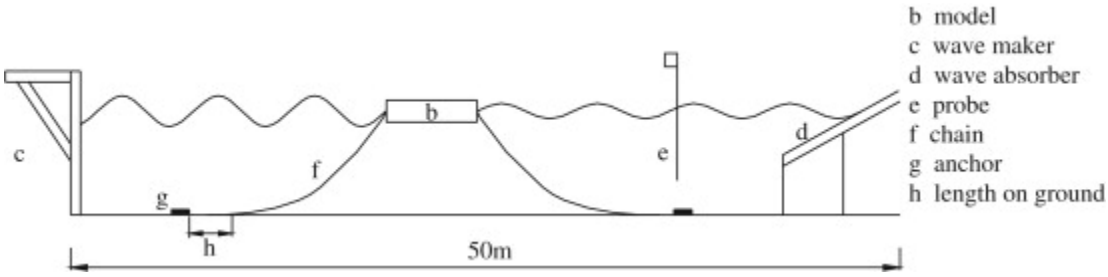


Fig. 12. Floating Breakwater using Tyre as model (b)



Fig. 13. Map of Sungei Buloh Nature Park Network

Appendix 1: Birds recorded at Sungei Buloh Wetland Reserve, Kranji Marshes and Sungei Buloh Nature Park Network as of 30 June 2023 as compared to all birds recorded in Singapore.

Number	Common Name	Scientific Name	Migration Status	Global Conservation Status (2023)	National Conservation Status (2021)	SBWR	KM	SBNPN
1	Wandering Whistling Duck	*Dendrocygna arcuata*	R	LC	NA	1	1	1
2	Lesser Whistling Duck	*Dendrocygna javanica*	R	LC	EN	1	1	1
3	Cotton Pygmy Goose	*Nettapus coromandelianus*	R	LC	CR		1	1
4	Garganey	*Anas querquedula*	M	LC	VU	1		1
5	Northern Shoveler	*Anas clypeata*	M	LC	NE	1		1
6	Gadwall	*Anas strepera*	M	LC	NE	1		1
7	Eurasian Wigeon	*Anas penelope*	M	LC	NE	1	1	1
8	Northern Pintail	*Anas acuta*	M	LC	NE	1		1
9	Tufted Duck	*Aythya fuligula*	M	LC	NE			
10	Red Junglefowl	*Gallus gallus*	R	LC	NT	1	1	1
11	King Quail	*Excalfactoria chinensis*	R	LC	LC	1	1	1
12	Malaysian Eared Nightjar	*Lyncornis temminckii*	R	LC	CR			
13	Grey Nightjar	*Caprimulgus jotaka*	M	LC	LC			
14	Large-tailed Nightjar	*Caprimulgus macrurus*	R	LC	LC	1	1	1
15	Savanna Nightjar	*Caprimulgus affinis*	R	LC	LC	1	1	1

Appendix 1 *(Cont'd)*

415	Eastern Yellow Wagtail	*Motacilla tschutschensis*	M	LC	VU	**1**	**1**	**1**
416	Citrine Wagtail	*Motacilla citreola*	M	LC	NE			**1**
417	Grey Wagtail	*Motacilla cinerea*	M	LC	LC	**1**	**1**	**1**
418	White Wagtail	*Motacilla alba*	M	LC	LC	**1**	**1**	**1**
419	Paddyfield Pipit	*Anthus rufulus*	R	LC	LC	**1**	**1**	**1**
420	Tree Pipit	*Anthus trivialis*	M	LC	NE			
421	Olive-backed Pipit	*Anthus hodgsoni*	M	LC	NE			
422	Red-throated Pipit	*Anthus cervinus*	M	LC	LC		**1**	**1**
	Total					**281**	**235**	**298**

Notes:

This 2023 Singapore Species list was shared by the Nature Society (Singapore) Bird Group.

Key:

R – resident, NBV – Non-breeding visitor, M – Migrant.

LC – Least Concern, NT – Near Threatened, VU – Vulnerable, EN – Endangered, CR – Critically Endangered, DD – Data Deficient, NA – Not Applicable, NE – Not Evaluated.

Fig. 14. Birds recorded in Sungei Buloh Wetland Reserve, Kranji Marshes, and Sungei Buloh Nature Park Network as of 30 June 2023.

Table 5. Comparison of materials

Materials	Reference	Sustainable
Tetrapot	Portland Cement, Aggregate, Steel Reinforcement (sometimes)	Low-Carbon Concrete, Recycled Aggregates, Natural Fibres for Reinforcement
Floating Breakwater	Conventional concrete, Steel, EPS or polyethylene floats, Timber	Geopolymer or low-carbon concrete, Recycled metal, Bio-based or recycled plastic floats, Sustainably sourced timber
Bio Revetment	Concrete Blocks or Gabions, Concrete Blocks or Gabions, Steel, Non-Biodegradable Geotextiles	Coir (Coconut Fiber), Bamboo, Recycled Plastic Composites, Biodegradable Geotextiles

Table 6. Comparison of environment impact between common and sustainable materials in Northwest Coast 1

	Common Material		Sustainable Material	
	Tetrapod	**Floating Breakwater**	**Tetrapod**	**Floating Breakwater**
Abiotic Depletion Potential	0.000015	0.12	0.075	0.045
Global Warming Potential	0.095	0.1925	0.0125	0.05
Ozone Depletion Potential	0.000015	0.0000047	0.000017	0.0000011
Acidification Potential	0.02	0.18	0.01	0.04
Photochemical Ozone Creation Potential	0.005	0.007	0.003	0.002
Eutrophication Potential	0.0025	0.0225	0.01	0.0045
Eco-cost of Materials (USD/Kg)	0.125	0.522	0.11	0.142

Table 7. Comparison of environment impact between common and sustainable materials in Northwest Coast 2

	Common Material		Sustainable Material	
	Tetrapod	**Floating Breakwater**	**Tetrapod**	**Floating Breakwater**
Abiotic Depletion Potential	0.000015	0.12	0.075	0.045
Global Warming Potential	0.095	0.1925	0.0125	0.05
Ozone Depletion Potential	0.000015	0.0000047	0.000017	0.0000011
Acidification Potential	0.02	0.18	0.01	0.04
Photochemical Ozone Creation Potential	0.005	0.007	0.003	0.002
Eutrophication Potential	0.0025	0.0225	0.01	0.0045
Eco-cost of Materials (USD/Kg)	0.125	0.522	0.11	0.142

Table 8. Comparison of environment impact between common and sustainable materials in Northwest Coast 2

	Common Material	**Sustainable Material**
	Bio Revetment	**Bio Revetment**
Abiotic Depletion Potential	0.045	0.01
Global Warming Potential	0.05	0.01
Ozone Depletion Potential	0.0000011	0.00000022
Acidification Potential	0.04	0.02
Photochemical Ozone Creation Potential	0.002	0.001
Eutrophication Potential	0.0045	0.003
Eco-cost of Materials (USD/Kg)	0.142	0.044

Table 9. Comparison of service life between common and sustainable materials [16]

	Reference	**Sustainable**
Tetrapod	100 years	97 years
Floating Breakwater	100 years	97 years
Bio Revetment	30 years	20 years

Table 10. Comparison of performance between common and sustainable materials [17]

	Reference	**Sustainable**
Tetrapod	52MPa	53MPa
Floating Breakwater	52MPa	53MPa
Bio Revetment	-	-